GIRISH SHARMA

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Education

National Institute of Technology Karnataka *Aug'23 – Present*

Bachelor of Technology in Computer Science *GPA: 7.70*

Work Experience

Software Engineering Intern (SDE) — Tech Mahindra *May'26 – July'26*

- Architected an end-to-end **AI-driven observability platform** using **FastAPI** and **Kafka**, automating log parsing, real-time anomaly detection, and incident remediation across distributed systems.
- Engineered an autonomous **Root-Cause Analysis (RCA)** engine with **LangGraph** multi-agent orchestration and a **Neo4j** knowledge graph, slashing MTTR from **40+ minutes to seconds** across **1.12M+** requests (**87%** confidence).
- Developed a self-monitoring telemetry framework with **14 real-time health detectors** and secure **data masking-demasking modules** to monitor LLM gateways, vector indices, and relational databases.
- Built a **WireMock** service-virtualization suite of **43 calibrated stubs** from **1.5M+** production logs, achieving zero-risk CI pipeline testing under loads of **400 requests/sec** with a **94.6%** pass rate across **160k+** checks.

Technologies Used: LangGraph, FastAPI, Kafka, Neo4j, OpenSearch, WireMock, Docker, Multi-Agent Systems, CI/CD, RCA Automation

Projects

TraceID - *Live Site* - Next.js, Python Flask, PostgreSQL, FaceNet, MTCNN *July'26 - Aug'26*

- Architected **TraceID**, a full-stack investigative platform that cross-references field photography against missing-person dossiers using multi-box face detection and **vector similarity** matching.
- Connected a Next.js frontend to Python Flask microservices for real-time **MTCNN** facial landmarking and **512-dim** FaceNet embeddings against a **Neon PostgreSQL** database.
- Achieved **>96%** candidate retrieval accuracy and reduced search latency to **<120ms** per query through **L2-normalized feature vector indexing**.

Meetlnk - *Live Site* - FastAPI, Python, LangChain, Pinecone, OpenAI API, Next.js *Aug'26 - Sept'26*

- Architected **Meetlnk**, an end-to-end meeting intelligence platform leveraging **Retrieval-Augmented Generation (RAG)** to parse long-form conversation transcripts and extract timestamped action items.
- Implemented speaker-turn sliding window chunking and **hybrid retrieval** combining **BM25** sparse lexical search with dense vector similarity over a **Pinecone** vector index.
- Integrated cross-encoder re-ranking to prune retrieval noise, cutting context token overhead by **42%** and lifting factual answer relevancy to **>94%** on semantic question-answering benchmarks.

BidBlitz - *Live Site* - Node.js, Express.js, PostgreSQL, JWT, bcrypt *Feb'25 - April'25*

- Architected a high-concurrency auction engine using **Node.js**, **Express**, and **PostgreSQL**, enabling sub-200ms bid updates and supporting 1,000+ concurrent simulated requests.
- Engineered a secure authentication pipeline with **JWT** and **bcrypt** salted hashing, mitigating unauthorized access and reducing session vulnerability by 100%.
- Designed a normalized **relational database schema** with optimized indexing and foreign key constraints, reducing average query retrieval time by 35%.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++, HTML5/CSS3, SQL

Frameworks & Technologies: Next.js, React.js, Node.js, Express.js, Flask, FastAPI, Tailwind CSS, REST APIs, FaceNet, MTCNN, OpenCV, LangGraph, Kafka, OpenSearch, Neo4j, Docker, JWT Authentication

Databases & Developer Tools: MongoDB, Mongoose, PostgreSQL, MySQL, Pinecone (Vector DB), Cloudinary, Git, GitHub, GitHub Copilot, Postman, VS Code, Jupyter Lab, Poetry

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Object Oriented Programming, Database Management Systems, Computer Architecture, Discrete Mathematics, Design of Digital Systems, Linear Algebra, Probability Theory and Statistics

Soft Skills: Effective Communication, Public Speaking, Teamwork, Problem Solving, Project Management, Adaptability

Achievements

- Secured an All India Rank of **1883** in **JEE Mains 2023** out of more than 1 million candidates.
- Achieved an All India Rank of **2616** in **JEE Advanced 2023** among top aspirants nationwide.

Leadership & Extracurricular Activities

Yoga Club, Co-Convener *Feb'24 – Feb'26*

- Oversaw planning and execution of club events and workshops, and managed engagement for 150+ active members.

SPICMACAY, Event Planner and Publicity Coordinator *March'24 – Present*

- Promoted awareness and appreciation of India's rich cultural heritage through diverse performances and workshops.